

Research briefing | UK

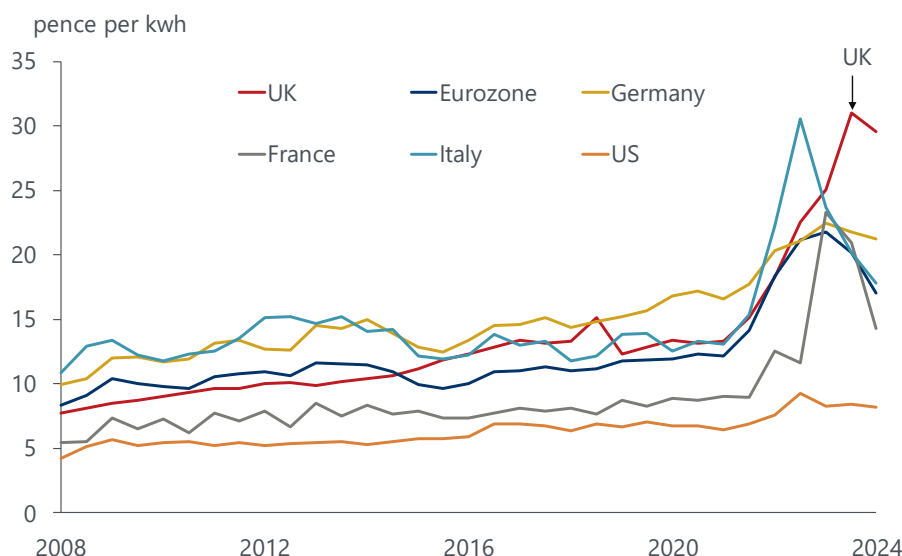
Why the energy shock persisted for firms, but may soon fade

- The energy price shock for UK firms – much worse than that for UK consumers, or for firms in the EU – may be finally beginning to fade.
- Average gas and electricity prices for UK firms fell slightly in both Q1 and Q2 this year. Even so, the cumulative increases in energy prices for UK firms over the last few years have been much greater than those for households, and far higher than the average for firms in the EU. In fact, average electricity prices for UK firms were the highest among major advanced economies last year and have probably remained so this year.
- The disparity between firms' energy costs in the UK and EU countries largely reflects the greater role of gas prices in setting electricity prices in the UK, which meant the 2022-2023 surge in gas prices caused a much greater rise in firms' overall energy prices in the UK. The longer persistence of the energy shock for UK firms than households is mainly due to the more widespread prevalence of fixed-price energy contracts among firms.
- More broadly, the indirect effects on prices from the surge in firms' energy costs has contributed to UK inflation persistence, especially in services, while also weakening activity, especially in energy-intensive sectors.
- There are signs that energy prices for UK firms are likely to fall further, as the lagged effects of recent declines in wholesale energy costs feed through. As this happens, the indirect effects of lower energy costs will help reduce core inflation, especially in services, and lend support to economic growth over the next year or two.

For households, the worst of the energy price shock passed some time ago, with household gas and electricity bills falling significantly during 2023 and early this year. However, for UK firms, the energy price shock was much larger and more persistent: prices rose further in 2023, and only fell slightly in H1 this year. Moreover, the average prices of electricity and gas for firms have risen further in the UK than the EU average, with UK firms' electricity costs now the highest among advanced economies.

Chart 1: UK firms face highest electricity costs among advanced economies

Selected countries: Average electricity costs (incl tax) for firms



Sources: Oxford Economics/Haver Analytics

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The energy price shock hit UK firms relatively hard

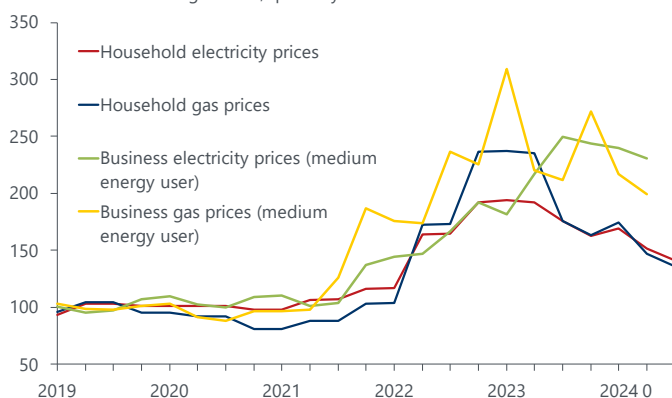
In the initial stages of the 2022-2023 energy price shock, UK households and firms both faced similarly large increases in gas and electricity prices. For example, in Q4 2022, average gas prices were 92% above the 2019 level for both households and firms. Electricity prices were 137% up for households, and 125% up for firms.

Since then, the energy price shock has proved much more persistent for firms than households. For example, in Q2 this year (latest comparable data), average gas prices for UK firms (with medium energy use) were 100% above their 2019 level, with average electricity prices 130% up, versus increases for households of 47% for gas and 52% for electricity.

Chart 2: The energy shock has been larger and more persistent for firms than for households

UK: Average energy prices for households and businesses

Indexed to 2019 average = 100, quarterly



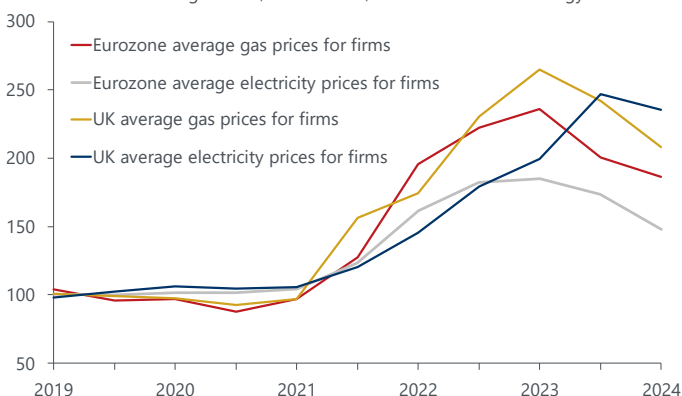
Sources: Oxford Economics/Haver Analytics

The energy price shock for UK firms has also been relatively large and persistent by international standards, especially for electricity prices. Data for EU firms' energy prices are only available for half-yearly periods, and on a comparable basis (for medium energy users), average gas prices for firms in H1 this year were up by 87% in the eurozone and 108% in the UK from their 2019 levels. For firms' average electricity prices, the increases were up by 48% in the eurozone and 135% in the UK.

Chart 3: The energy shock for firms has also been more persistent in the UK than the eurozone

UK and eurozone: Trends in firms' energy prices

Indexed to 2019 average = 100, semi-annual, firms with medium energy use



Sources: Oxford Economics/Haver Analytics

In levels terms, using comparable Eurostat data converted to sterling, UK firms' gas prices are not far from the eurozone average. We estimate that from 2008-2019 UK firms' electricity prices (including tax) on average were also similar to the eurozone average, but in H1 this year they were about 75% above it, the widest disparity on record. We estimate that in H1 this year the average level of electricity prices for UK

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firms exceeded those in any EU country – about 40% above levels in Germany, double that in France, and more than three times the US level.

[Official figures](#) show that in 2023, UK firms' average electricity prices were the highest of any advanced economy. Given the available data, the UK probably remained top of this list in the first half of this year (**Chart 1**).

The UK's high level of electricity costs is especially important for firms, because electricity plays a much bigger role in total energy costs for firms than for households. For example, over 2013-2022, UK households on average spent 27% more on electricity than gas, whereas UK firms spent 380% more on electricity than gas. So electricity costs play the main role in UK firms' overall energy costs.

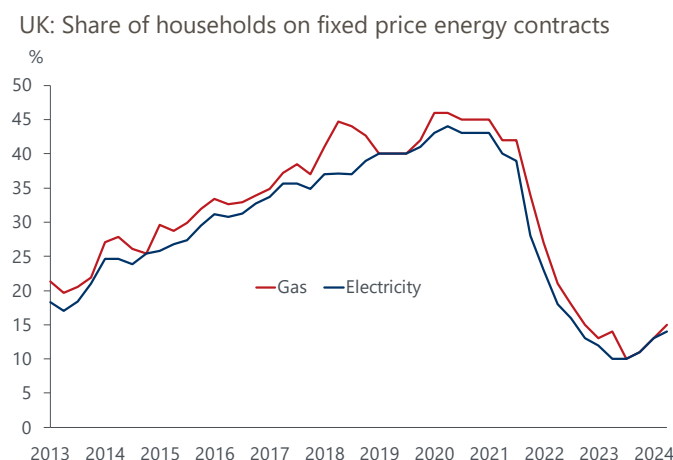
Why has this happened?

The disparity between trends in energy costs for UK households and firms mainly reflects the differing extent of official support measures, which benefited households much more than firms.

There was a direct effect, because [the energy price cap](#) put a ceiling on household energy prices, especially in early 2023. For firms, there were some [energy support schemes](#) but not a general price cap. As a result, the surge in wholesale energy costs fed through much more powerfully for firms than households.

There was also an indirect behavioural effect. With the government shielding them from further upside in energy costs, many households judged they could afford to take the gamble of shifting to variable rate tariffs. The [share of households on fixed rate tariffs](#) (which generally keep prices fixed for 12-24 months) fell from 43% for electricity and 45% for gas in March 2021 to 16% and 18% respectively in September 2022, and remained similarly low (14% and 15%, respectively) in June this year. So households have benefited relatively quickly from recent declines in wholesale prices.

Chart 4: Many households have shifted to variable rate tariffs



Sources: Oxford Economics/Haver Analytics

By contrast, it appears that even when energy prices were very high, firms valued the safety of a fixed price contract to protect them against prices going even higher, perhaps because they judged that if energy prices did rise, government support would again be focussed elsewhere. [ONS figures](#) show that in September 2022, the share of firms (excluding micro-firms with less than 10 employees) on fixed price contracts for gas and electricity was 49% and 53% respectively, with 20% and 18% respectively on variable price contracts (the rest did not know). Separate [figures](#) also show that in late-2022, 51% of small firms were on fixed-price energy contracts. In September this year, the UK government [noted](#) that "Many businesses are on fixed price contracts". Moreover, fixed energy contracts for firms [tend to be longer than those for households](#), ranging between one and five years.

The use of fixed price contracts initially delayed the impact of higher energy costs on UK firms. But the upshot is that many firms this year have still been locked into relatively expensive energy price contracts

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that were fixed at the relatively high wholesale prices of 2022-2023, and have been slow to benefit from recent declines in wholesale prices for gas and electricity.

Why are UK electricity prices so high compared to EU countries?

A limited part of the disparity between firms' electricity costs in the UK and elsewhere in Europe reflects taxes: in H2 2023, UK firms' electricity prices on a comparable basis were 78% above the EU median including tax, and 71% above it excluding tax. UK prices are still very high by international standards excluding tax.

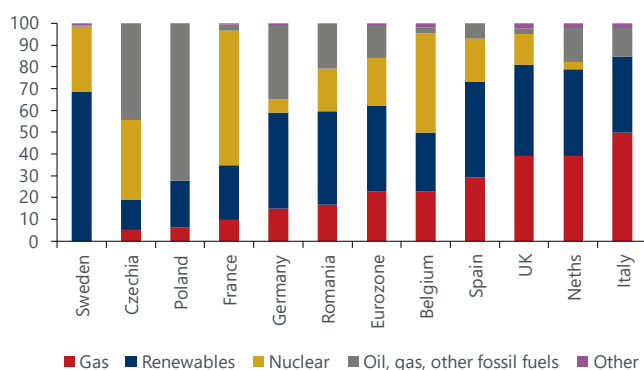
The main cause of this is the bigger role of natural gas prices in setting UK electricity prices. To understand this, we need to delve into the workings of the UK electricity market. Wholesale electricity prices in the UK and much of Europe are determined in [an auction process](#) by the marginal generation cost of the highest-cost producer in each country needed to meet demand. The marginal producer of electricity in the UK is most often gas-powered because gas plays a relatively big role in UK electricity generation and is also one of the most expensive sources. Renewables and nuclear [produce most UK electricity](#) but have relatively low marginal cost and are usually not large enough to meet demand.

As a result, the price that is paid for electricity in wholesale markets (and hence eventually for firms) is usually set much higher, at the marginal cost of generating electricity with gas. By contrast, EU countries typically make much more use of nuclear power as well as coal, oil and other fossil fuels.

Chart 5: Gas plays a relatively large role in UK electricity generation

UK and EU Countries: Electricity generation by source of power

% of total



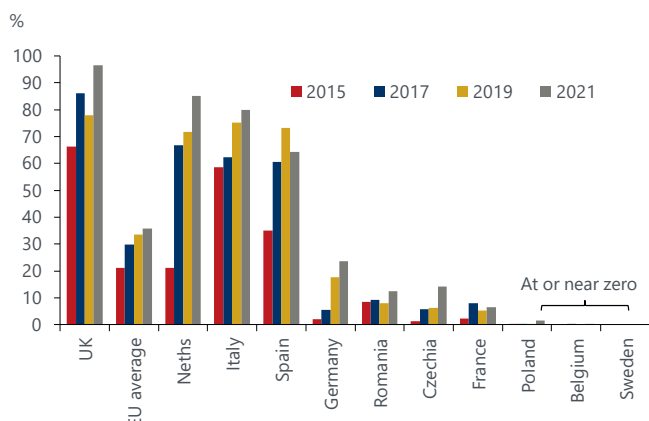
Source: Oxford Economics / Haver Analytics

A [recent study](#) found that in 2021 (latest data) gas-powered generation costs set wholesale electricity prices in the UK for 97% of the time - higher than in any EU country, and compared to an average of 36% of the time in EU countries (for which solid fossil fuels, nuclear and renewables all play a much bigger role than in the UK). For example, in recent years, the most frequent driver of wholesale gas prices has been coal and solid fossil fuels in Germany, nuclear in France, and hydro in Sweden. As a result, the 2022-2023 surge in gas prices (and its impact on contracted prices for gas-powered electricity generation) caused greater upward pressure on electricity prices in the UK than elsewhere.

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Chart 6: Gas plays a much bigger role in setting UK wholesale electricity prices than elsewhere

UK & EU: Pct of time gas generation costs set electricity prices



Source: Oxford Economics / Haver Analytics

Hence, although UK firms spend relatively little on direct use of gas, their energy costs are highly sensitive to gas prices. Moreover, the link to gas prices introduces an extra lag which has increased the persistence of the energy shock in the UK. This is because changes in wholesale gas prices feed through to costs of gas-powered electricity generators with a lag because of generators' fixed price contracts for gas. And there is a further lag between changes in wholesale electricity prices and firms' energy costs because firms also have fixed price contracts for electricity supply.

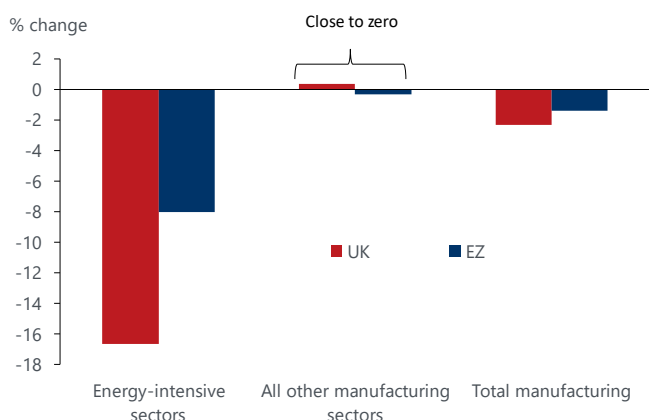
The effects of the energy price shock have been large and persistent in the UK

The energy price surge has been an adverse supply shock, which added to inflation and reduced real activity. The greater scale and persistence of this shock in the UK, especially for firms, means these effects have been relatively large and enduring. In competitive global markets, UK firms have been at a significant disadvantage.

For example, [energy-intensive manufacturing output](#) (defined as paper, refinery products, chemicals, non-metallic minerals, and basic metals) has fallen sharply, down by 17% between H2 2019 and Q3 2024 in the UK compared to declines of 14% in Germany and 8% in the eurozone. By contrast, output in other parts of manufacturing was little changed over this period in both the UK and eurozone.

Chart 7: Energy-intensive manufacturing output has fallen especially sharply in the UK

UK and EZ: Change in manufacturing output since H2-2019



Source: Oxford Economics / Haver Analytics

Similarly, the UK's trade deficit in energy-intensive manufacturing (excluding basic metals, which is distorted by trade in precious metals) has risen from £16.2bn in 2019 to an annualised rate of £32.7bn in the first three quarters of this year. Exports have fallen in nominal terms despite rising prices, and imports have risen markedly as UK firms source energy-intensive products from elsewhere.

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And, of course, the drag on growth from high energy costs has been felt in other sectors as well.

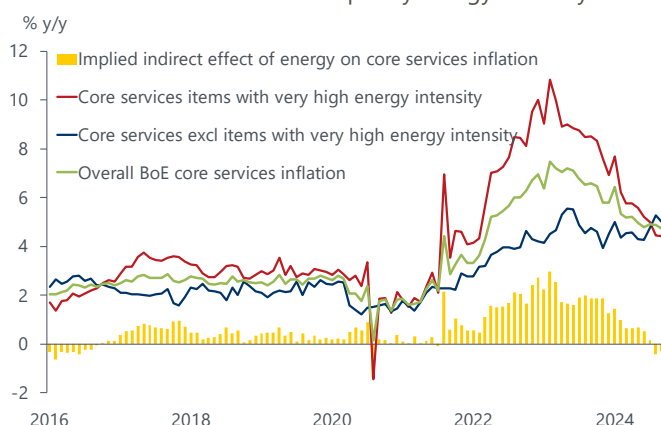
Moreover, as we have [highlighted previously](#), indirect effects of the energy price surge caused UK inflation to rise sharply in the more energy-intensive components of the CPI, adding significantly to overall inflation in 2022-2023. These effects were especially marked for services inflation in the UK, because CPI services items are [more energy-intensive](#) than CPI core goods.

These indirect effects from energy prices are also large in the Bank of England's [preferred core services inflation measure](#), which excludes indexed and volatile components, rents, and foreign holidays. This series excludes many items with relatively low energy intensity, and many of those that are included are relatively energy-intensive, e.g. restaurants, cafes, takeaway food, and canteens, which jointly account for 45% of the BoE's core services measure. In all, 52% (by CPI weight) of the BoE's core services measure is [classified by the ONS](#) as having "very high" energy intensity, compared to 23% of all items in the normal core CPI measure, which excludes food, drink, alcohol, and tobacco.

The recent swings in the BoE's core services inflation measure – up during 2022-2023 and down since then – have largely been concentrated in items with "very high" energy intensity. Excluding items with very high energy intensity, the BoE's core services measure did not rise as much in 2022-2023, but has been roughly stable at 4%-5% y/y since then.

Chart 8: Indirect effects from energy prices have affected the BoE's core services gauge

UK: BoE core services inflation split by energy intensity



Sources: Oxford Economics/Haver Analytics

The gap between overall BoE core services measure and the inflation rate of core services excluding items with very high energy intensity (i.e. the gap between the green and blue lines in [Chart 8](#)) represents a rough guide to the indirect effects of energy prices on this core inflation gauge. This estimate peaked at 2ppt-3ppt early last year and has recently turned slightly negative – i.e. the inflation rate of energy-intensive items is now slightly below overall core services inflation.

Of course, this is only a rough estimate: other factors may have lifted the inflation rate of energy-intensive core services and increases in energy prices also have probably lifted prices for items that are not very energy-intensive. But a key point is that indirect effects from energy prices have probably played a big role in so-called core inflation gauges. Moreover, with firms' energy costs having fallen only slightly so far, the previous surge in inflation among energy-intensive sectors has not yet been followed by a corresponding period of very low inflation in these sectors. And, so far, there is not much sign that the recent slowdown in pay growth has fed through to core services inflation.

In the near-term, lower energy costs will help lift real growth and cap inflation

In the near-term, lagged indirect effects of the recent modest declines in firms' energy costs will continue to lower core services inflation. Moreover, unless wholesale energy costs rise sharply, firms' energy costs are likely to fall further as recent declines in wholesale energy prices work through. For example, the average cost of gas for gas-powered electricity generators fell by 60% between Q4 last year and Q2 this year. As this began to feed through to firms (with another lag due to firms' fixed price contracts), UK firms' average

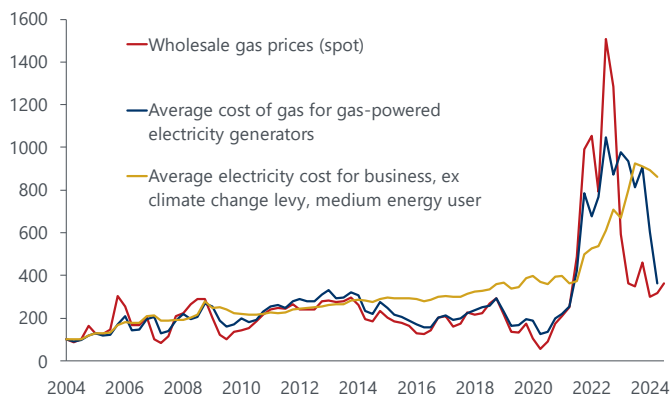
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electricity costs fell in Q2. And unless wholesale gas prices rise sharply, firms' average energy costs are likely to fall further in coming quarters.

Chart 9: With a lag, lower wholesale gas prices should help bring firms' electricity costs lower

UK: Wholesale gas prices and firms' electricity prices

Indexed to Q1 2004 = 100



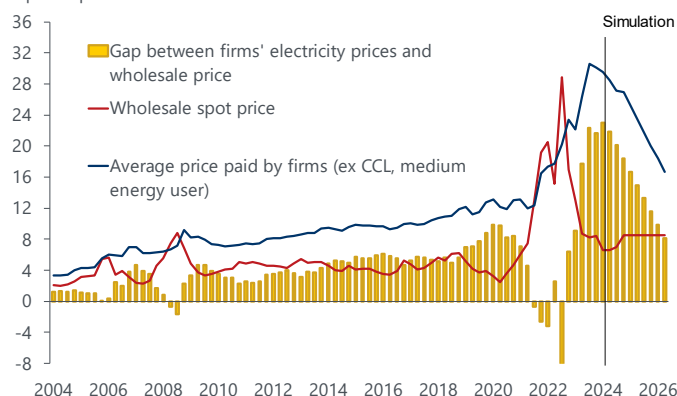
Sources: Oxford Economics/Haver Analytics

As a rough guide, we have estimated a scenario in which wholesale energy costs remain at their average level in October this year (which was slightly above levels in Q2 and Q3), while the spread between wholesale energy costs and the average prices charged to UK firms (excluding the climate change levy) gradually returns to the 2018-2019 average (adjusted for inflation) by the middle of 2026. This spread reflects distributors' costs and profits, as well as effects due to fixed price contracts. In this scenario, UK firms' average cost of gas will fall by about 8% and average electricity costs will fall by about 40% between Q2 this year and mid-2026 (see **Chart 10** for the implied path of electricity prices for a firm with medium energy use).

Chart 10: If spreads over wholesale prices normalise, firms' average electricity costs could fall about 40%

UK: Simulation for firms' average electricity prices

pence per kwh



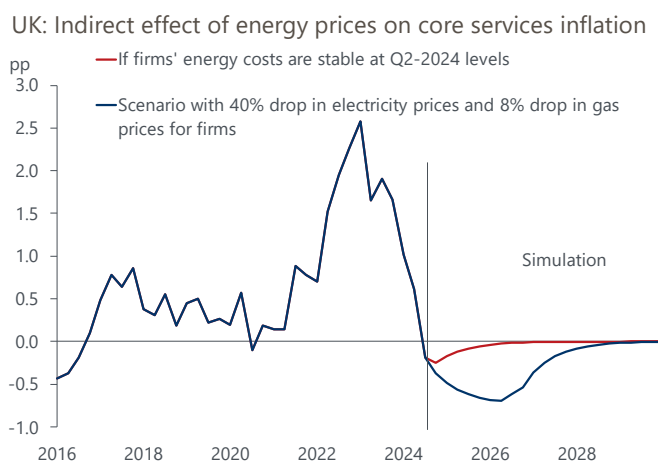
Sources: Oxford Economics/Haver Analytics

Note that in this simulation, the declines in firms' average energy costs reflect the delayed impact (as fixed price contracts expire and are reset) from declines in wholesale prices that already occurred, with no further changes in wholesale prices. Some firms – those on contracts with variable energy prices, or prices that reset annually – would already have experienced these price declines, while some of those on fixed price contracts would see bigger falls.

Based on recent experience, we estimate that the indirect effects of these declines in firms' energy costs would cut the BoE's core services inflation measure by about 0.6ppts in both 2025 and 2026, with a smaller downward effect in 2027. Almost all this effect comes from lower electricity prices.

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Chart 11: Lower energy costs for firms could help dampen core services inflation significantly



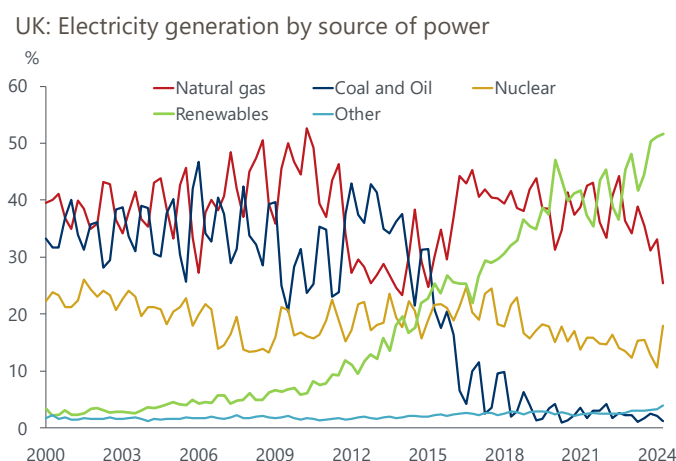
Source: Oxford Economics / Haver Analytics

We stress this is a highly uncertain estimate. But rather than the precise figure, the key point is that a sizeable disinflationary effect for services inflation (and core inflation) may lie ahead from the lagged impact of declines in wholesale energy costs over the last year or two. And this could be reinforced by the disinflationary effects of lower pay growth (which do not seem to have played a big role so far).

The link from gas to electricity prices likely to weaken further ahead

Over time, the link from gas prices to UK electricity prices is likely to weaken as the rising role of renewables reduces the share of gas in UK electricity production. In Q2 this year (latest data), the share of UK gas-powered generation in UK electricity supply fell to 25% from 43% three years earlier, with the renewables share up from 37% to 52% over the same period. Partial data, covering major electricity generators, suggests the role of gas in UK electricity generation weakened further in July-August.

Chart 12: The role of gas in UK electricity production has recently fallen sharply



Sources: Oxford Economics/Haver Analytics

If this trend continues then, over time, the role of gas prices in setting wholesale electricity prices is likely to diminish, because, as energy use fluctuates, there will be more occasions on which gas is not the marginal source of supply. In turn, that would make the UK less vulnerable to future swings in gas prices, reducing the UK's current heightened sensitivity.